

Module: Notifications

[pawguard/modules/notifications](#) — API prefix: `/notifications`

PawGuard Backend | FastAPI 3.13 + PostgreSQL 17 + Redis/ARQ + S3 | Generated from repo @ commit 99a8430 (2026-08-12)

1. Purpose

Shared delivery layer for in-app, email, and push notifications, consumed by nearly every other module (reminders, broadcasts, approvals, alerts).

2. Position in the Layered Architecture

[Client](#) → [Router \(auth+validate\)](#) → [Service \(business logic\)](#) → [Repository \(data access\)](#) → [PostgreSQL](#)

Like every module in this codebase, business rules live only in the Service layer; Routers authenticate/authorize/validate and Repositories perform data access with no business decisions (see AGENTS.md — the project's engineering constitution).

3. Data Model — Owned Tables

Model Class	Table Name
Notification	notifications
NotificationPreference	notification_preferences

4. API Endpoints (10 total)

Method	Path	Permission Required
GET	/notifications	—
GET	/notifications/unread-count	—
PUT	/notifications/{notification_id}/read	—
PUT	/notifications/read-all	—
DELETE	/notifications/{notification_id}	—
POST	/notifications/bulk/delete	notification:manage
POST	/notifications/send	notification:manage
GET	/notifications/preferences	—
PUT	/notifications/preferences	—
POST	/notifications/broadcast	system:admin

Distinct permission codes used: notification:manage, system:admin

5. Key Service-Layer Operations

- broadcast
- broadcast_to_all_users
- bulk_delete
- count_unread
- create_notification
- delete_notification
- get_preferences
- list_paginated
- mark_all_read
- mark_read

- `send_notification`
- `update_preferences`

6. Notable Implementation Details

- Consumed by ARQ background jobs (reminders, broadcasts) as well as inline request handlers.
- Delivery channels: in-app + transactional email (Brevo HTTP API); push is not yet wired to a device-token provider.

Auto-extracted from `pawguard/modules/notifications/` (`models.py`, `router(s)`, `service.py`) at repo commit 99a8430, 2026-08-12.